2 The hydrogen atom

The simple hydrogen atom has had a great influence on the development of quantum theory, particularly in the first half of the twentieth century when the foundations of quantum mechanics were laid. As measurement techniques improved, finer and finer details were resolved in the spectrum of hydrogen until eventually splittings of the lines were observed that cannot be explained even by the fully relativistic formulation of quantum mechanics, but require the more advanced theory of quantum electrodynamics. In the first chapter we looked at the Bohr–Sommerfeld theory of hydrogen that treated the electron orbits classically and imposed quantisation rules upon them. This theory accounted for many of the features of hydrogen but it fails to provide a realistic description of systems with more than one electron, e.g. the helium atom. Although the simple picture of electrons orbiting the nucleus, like planets round the sun, can explain some phenomena, it has been superseded by the Schrödinger equation and wavefunctions. This chapter outlines the application of this approach to solve Schrödinger's equation for the hydrogen atom; this leads to the same energy levels as the Bohr model but the wavefunctions give much more information, e.g. they allow the rates of the transitions between levels to be calculated (see Chapter 7). This chapter also shows how the perturbations caused by relativistic effects lead to fine structure.

2.1 The Schrödinger equation

The solution of the Schrödinger equation for a Coulomb potential is in every quantum mechanics textbook and only a brief outline is given here.[1] The Schrödinger equation for an electron of mass m_e in a spherically-symmetric potential is

$$\left\{\frac{-\hbar^2}{2m_e}\nabla^2 + V(r)\right\}\psi = E\psi\,. \tag{2.1}$$

This is the quantum mechanical counterpart of the classical equation for the conservation of total energy expressed as the sum of kinetic and potential energies.[2] In spherical polar coordinates we have

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{1}{r^2}\mathbf{l}^2\,, \tag{2.2}$$

[1] The emphasis is on the properties of the wavefunctions rather than how to solve differential equations.

[2] The operator for linear momentum is $\mathbf{p} = -\mathrm{i}\hbar\boldsymbol{\nabla}$ and for angular momentum it is $\hbar\mathbf{l} = \mathbf{r}\times\mathbf{p}$. This notation differs in two ways from that commonly used in quantum texts. Firstly, $\hbar$ is taken outside the angular momentum operators, and secondly, the operators are written without 'hats'. This is convenient for atomic physics, e.g. in the vector model for the addition of angular momenta.

where the operator $\mathbf{l}^2$ contains the terms that depend on θ and ϕ, namely

$$\mathbf{l}^2 = -\left\{ \frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \, \frac{\partial}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} \right\}, \tag{2.3}$$

and $\hbar^2 \mathbf{l}^2$ is the operator for the orbital angular momentum squared. Following the usual procedure for solving partial differential equations, we look for a solution in the form of a product of functions $\psi = R(r)Y(\theta, \phi)$. The equation separates into radial and angular parts as follows:

$$\frac{1}{R} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) - \frac{2m_{\mathrm{e}} r^2}{\hbar^2} \left\{ V(r) - E \right\} = \frac{1}{Y} \mathbf{l}^2 Y \, . \tag{2.4}$$

Each side depends on different variables and so the equation is only satisfied if both sides equal a constant that we call b. Thus

$$\mathbf{l}^2 Y = b \, Y \, . \tag{2.5}$$

This is an eigenvalue equation and we shall use the quantum theory of angular momentum operators to determine the eigenfunctions $Y(\theta, \phi)$.

2.1.1 Solution of the angular equation

To continue the separation of variables we substitute $Y = \Theta(\theta)\Phi(\phi)$ into eqn 2.5 to obtain

$$\frac{\sin\theta}{\Theta} \frac{\partial}{\partial\theta} \left(\sin\theta \, \frac{\partial\Theta}{\partial\theta} \right) + b \sin^2\theta = -\frac{1}{\Phi} \frac{\partial^2 \Phi}{\partial\phi^2} = \text{const} \, . \tag{2.6}$$

The equation for $\Phi(\phi)$ is the same as in simple harmonic motion, so[3]

$$\Phi = A \mathrm{e}^{\mathrm{i}m\phi} + B \mathrm{e}^{-\mathrm{i}m\phi} \, . \tag{2.7}$$

[3] A and B are arbitrary constants. Alternatively, the solutions can be written in terms of real functions as $A' \sin(m\phi) + B' \cos(m\phi)$.

The constant on the right-hand side of eqn 2.6 has the value m^2. Physically realistic wavefunctions have a unique value at each point and this imposes the condition $\Phi(\phi + 2\pi) = \Phi(\phi)$, so m must be an integer. The function $\Phi(\phi)$ is the sum of eigenfunctions of the operator for the z-component of orbital angular momentum

$$\hbar l_z = -\mathrm{i}\hbar \frac{\partial}{\partial\phi} \, . \tag{2.8}$$

The function $\mathrm{e}^{\mathrm{i}m\phi}$ has magnetic quantum number m and its complex conjugate $\mathrm{e}^{-\mathrm{i}m\phi}$ has magnetic quantum number $-m$.[4]

A convenient way to find the function $Y(\theta, \phi)$ and its eigenvalue b in eqn 2.5[5] is to use the ladder operators $l_+ = l_x + \mathrm{i}l_y$ and $l_- = l_x - \mathrm{i}l_y$. These operators commute with $\mathbf{l}^2$, the operator for the total angular momentum squared (because l_x and l_y commute with $\mathbf{l}^2$); therefore, the three functions Y, l_+Y and l_-Y are all eigenfunctions of $\mathbf{l}^2$ with the same eigenvalue b (if they are non-zero, as discussed below). The ladder

[4] The operator $-\partial^2/\partial\phi^2 \equiv l_z^2$ and consequently $\Phi(\phi)$ is an eigenfunction of l_z^2 with eigenvalue m^2.

[5] The solution of equations involving the angular part of ∇^2 arises in many situations with spherical symmetry, e.g. in electrostatics, and the same mathematical tools could be used here to determine the properties of the spherical harmonic functions, but angular momentum methods give more physical insight for atoms.

operators can be expressed in polar coordinates as:

$$
\begin{aligned}
l_+ &= \mathrm{e}^{\mathrm{i}\phi}\left(\frac{\partial}{\partial\theta} + \mathrm{i}\cot\theta\,\frac{\partial}{\partial\phi}\right), \\
l_- &= \mathrm{e}^{-\mathrm{i}\phi}\left(-\frac{\partial}{\partial\theta} + \mathrm{i}\cot\theta\,\frac{\partial}{\partial\phi}\right).
\end{aligned}
\tag{2.9}
$$

The operator l_+ transforms a function with magnetic quantum number m into another angular momentum eigenfunction that has eigenvalue $m+1$. Thus l_+ is called the raising operator.[6] The lowering operator l_- changes the magnetic quantum number in the other direction, $m \to m-1$. It is straightforward to prove these statements and other properties of these operators;[7] however, the purpose of this section is not to present the general theory of angular momentum but simply to outline how to find the eigenfunctions (of the angular part) of the Schrödinger equation.

Repeated application of the raising operator does not increase m indefinitely—for each eigenvalue b there is a maximum value of the magnetic quantum number[8] that we shall call l, i.e. $m_{\rm max} = l$. The raising operator acting on an eigenfunction with $m_{\rm max}$ gives zero since by definition there are no eigenfunctions with $m > m_{\rm max}$. Thus solving the equation $l_+Y = 0$ (Exercise 2.11) we find that the eigenfunctions with $m_{\rm max} = l$ have the form

$$Y \propto \sin^l\theta\,\mathrm{e}^{\mathrm{i}l\phi}\,. \tag{2.10}$$

Substitution back into eqn 2.5 shows that these are eigenfunctions $\mathbf{l}^2$ with eigenvalue $b = l(l+1)$, and l is the orbital angular momentum quantum number. The functions $Y_{l,m}(\theta,\phi)$ are labelled by their eigenvalues in the conventional way.[9] For $l = 0$ only $m = 0$ exists and $Y_{0,0}$ is a constant with no angular dependence. For $l = 1$ we can find the eigenfunctions by starting from the one with $l = 1 = m$ (in eqn 2.10) and using the lowering operator to find the others:

$$
\begin{aligned}
Y_{1,1} &\propto \sin\theta\,\mathrm{e}^{\mathrm{i}\phi}\,, \\
Y_{1,0} &\propto l_-Y_{1,1} \propto \cos\theta\,, \\
Y_{1,-1} &\propto l_-Y_{1,0} \propto \sin\theta\,\mathrm{e}^{-\mathrm{i}\phi}\,.
\end{aligned}
$$

This gives all three eigenfunctions expected for $l = 1$.[10] For $l = 2$ this procedure gives

$$
\begin{aligned}
Y_{2,2} &\propto \sin^2\theta\,\mathrm{e}^{\mathrm{i}2\phi}\,, \\
&\ \vdots \\
Y_{2,-2} &\propto \sin^2\theta\,\mathrm{e}^{-\mathrm{i}2\phi}\,.
\end{aligned}
$$

These are the five eigenfunctions with $m = 2, 1, 0, -1, -2$.[11] Normalised angular functions are given in Table 2.1.

Any angular momentum eigenstate can be found from eqn 2.10 by

[6] The raising operator contains the factor $\mathrm{e}^{\mathrm{i}\phi}$, so that when it acts on an eigenfunction of the form $Y \propto \Theta(\theta)\mathrm{e}^{\mathrm{i}m\phi}$ the resulting function l_+Y contains $\mathrm{e}^{\mathrm{i}(m+1)\phi}$. The θ-dependent part of this function is found below.

[7] These properties follow from the commutation relations for angular momentum operators (see Exercise 2.1).

[8] This statement can be proved rigorously using angular momentum operators, as shown in quantum mechanics texts.

[9] The dubious reader can easily check that $l_+Y_{l,l} = 0$. It is trivially obvious that $l_zY_{l,l} = l\,Y_{l,l}$, where $m = l$ for this function.

[10] $l_-Y_{1,-1} = 0$ and $m = -1$ is the lowest eigenvalue of l_z. Proportional signs have been used to avoid worrying about normalisation; this leaves an ambiguity about the relative phases of the eigenfunctions but we shall choose them in accordance with usual convention.

[11] The relation $Y_{l,-m} = Y^*_{l,m}$ shows that, if $m_{\rm max} = l$, then $m_{\rm min} = -l$. Between these two extremes there are $2l+1$ possible values of the magnetic quantum number m for each l. Note that the *orbital angular momentum quantum number* l is not the same as the length of the angular momentum vector (in units of $\hbar$). Quantum mechanics tells us only that the expectation value of the square of the orbital angular momentum is $l(l+1)$, in units of $\hbar^2$. The length itself does not have a well-defined value in quantum mechanics and it does not make sense to refer to it. When people say that an atom has 'orbital angular momentum of one, two, etc.', strictly speaking they mean that the orbital angular momentum quantum number l is 1, 2, etc.

Table 2.1 Orbital angular momentum eigenfunctions.

$$Y_{0,0} = \sqrt{\frac{1}{4\pi}}$$

$$Y_{1,0} = \sqrt{\frac{3}{4\pi}} \cos\theta$$

$$Y_{1,\pm1} = \mp\sqrt{\frac{3}{8\pi}} \sin\theta \, \mathrm{e}^{\pm \mathrm{i}\phi}$$

$$Y_{2,0} = \sqrt{\frac{5}{16\pi}} \left(3\cos^2\theta - 1\right)$$

$$Y_{2,\pm1} = \mp\sqrt{\frac{15}{8\pi}} \sin\theta \cos\theta \, \mathrm{e}^{\pm \mathrm{i}\phi}$$

$$Y_{2,\pm2} = \sqrt{\frac{15}{32\pi}} \sin^2\theta \, \mathrm{e}^{\pm 2\mathrm{i}\phi}$$

$$\text{Normalisation:} \int_0^{2\pi}\int_0^{\pi} |Y_{l,m}|^2 \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi = 1$$

repeated application of the lowering operator:[12]

$$Y_{l,m} \propto (l_-)^{l-m} \sin^l\theta \, \mathrm{e}^{\mathrm{i}l\phi} \, . \tag{2.11}$$

[12] This eigenfunction has magnetic quantum number $l - (l - m) = m$.

To understand the properties of atoms, it is important to know what the wavefunctions look like. The angular distribution needs to be multiplied by the radial distribution, calculated in the next section, to give the square of the wavefunction as

$$|\psi(r,\theta,\phi)|^2 = R_{n,l}^2(r) \, |Y_{l,m}(\theta,\phi)|^2 \, . \tag{2.12}$$

This is the probability distribution of the electron, or $-e\,|\psi|^2$ can be interpreted as the electronic charge distribution. Many atomic properties, however, depend mainly on the form of the angular distribution and Fig. 2.1 shows some plots of $|Y_{l,m}|^2$. The function $|Y_{0,0}|^2$ is spherically symmetric. The function $|Y_{1,0}|^2$ has two lobes along the z-axis. The squared modulus of the other two eigenfunctions of $l = 1$ is proportional to $\sin^2\theta$. As shown in Fig. 2.1(c), there is a correspondence between these distributions and the circular motion of the electron around the z-axis that we found as the normal modes in the classical theory of the Zeeman effect (in Chapter 1).[13] This can be seen in Cartesian coordinates where

$$\begin{aligned} Y_{1,0} &\propto \frac{z}{r} \, , \\ Y_{1,1} &\propto \frac{x + \mathrm{i}y}{r} \, , \\ Y_{1,-1} &\propto \frac{x - \mathrm{i}y}{r} \, . \end{aligned} \tag{2.13}$$

[13] Stationary states in quantum mechanics correspond to the time-averaged classical motion. In this case both directions of circular motion about the x-axis give the same distribution.

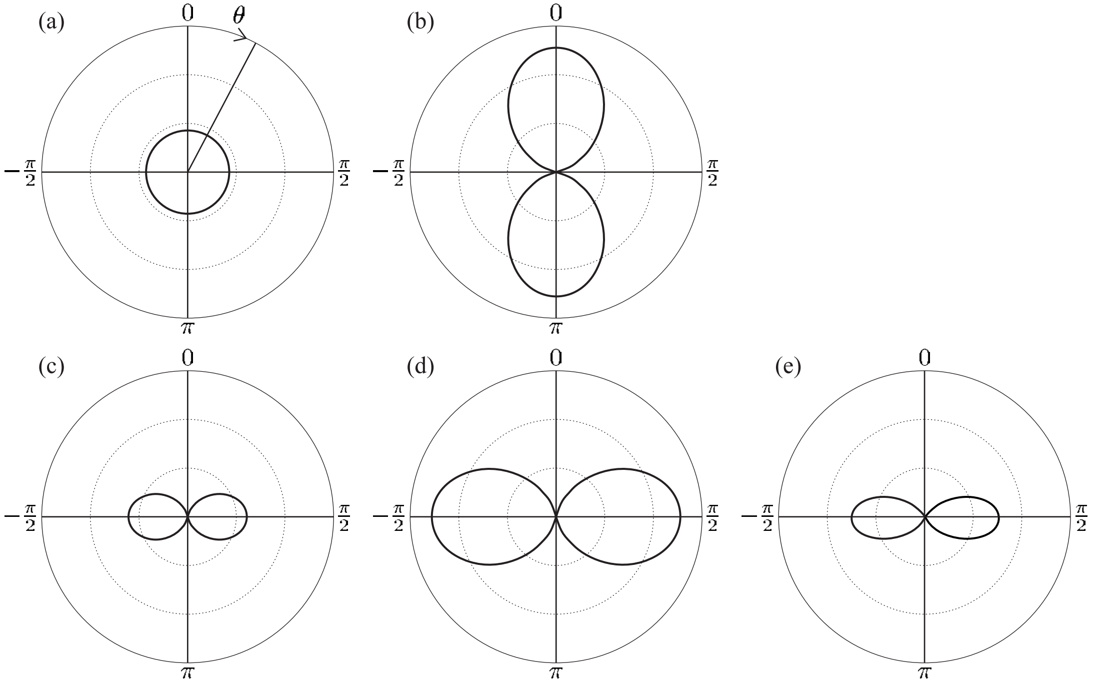

Fig. 2.1 Polar plots of the squared modulus of the angular wavefunctions for the hydrogen atom with $l = 0$ and 1. For each value of the polar angle θ a point is plotted at a distance proportional to $|Y(\theta, \phi)|^2$ from the origin. Except for (d), the plots have rotational symmetry about the z-axis and look the same for any value of ϕ. (a) $|Y_{0,0}|^2$ is spherical. (b) $|Y_{1,0}|^2 \propto \cos^2\theta$ has two lobes along the z-axis. (c) $|Y_{1,1}|^2 \propto \sin^2\theta$ has an 'almost' toroidal shape—this function equals zero for $\theta = 0$. ($|Y_{1,-1}|^2$ looks the same.) (d) $|Y_{1,1} - Y_{1,-1}|^2 \propto |x/r|^2$ has rotational symmetry about the x-axis and this polar plot is drawn for $\phi = 0$; it looks like (b) but rotated through an angle of $\pi/2$. (e) $|Y_{2,2}|^2 \propto \sin^4\theta$.

Any linear combination of these is also an eigenfunction of $\mathbf{l}^2$, e.g.

$$Y_{1,-1} - Y_{1,1} \propto \frac{x}{r} = \sin\theta\,\cos\phi\,, \tag{2.14}$$

$$Y_{1,-1} + Y_{1,1} \propto \frac{y}{r} = \sin\theta\,\sin\phi\,. \tag{2.15}$$

These two real functions have the same shape as $Y_{1,0} \propto z/r$ but are aligned along the x- and y-axes, respectively.[14] In chemistry these distributions for $l = 1$ are referred to as p-orbitals. Computer programs can produce plots of such functions from any desired viewing angle (see Blundell 2001, Fig. 3.1) that are helpful in visualising the functions with $l > 1$. (For $l = 0$ and 1 a cross-section of the functions in a plane that contains the symmetry axis suffices.)

[14] In the absence of an external field to break the spherical symmetry, all axes are equivalent, i.e. the atom does not have a preferred direction so there is symmetry between the x-, y- and z-directions. In an external magnetic field the states with different values of m (but the same l) are not degenerate and so linear combinations of them are not eigenstates of the system.

2.1.2 Solution of the radial equation

An equation for $R(r)$ is obtained by setting eqn 2.4 equal to the constant $b = l(l+1)$ and putting in the Coulomb potential $V(r) = -e^2/4\pi\epsilon_0 r$. It can be cast in a convenient form by the substitution $P(r) = rR(r)$:

$$-\frac{\hbar^2}{2m_e}\frac{d^2P}{dr^2} + \left\{\frac{\hbar^2}{2m_e}\frac{l(l+1)}{r^2} - \frac{e^2/4\pi\epsilon_0}{r} - E\right\}P = 0\,. \tag{2.16}$$

The term proportional to $l(l+1)/r^2$ is the kinetic energy associated with the angular degrees of freedom; it appears in this radial equation as an effective potential that tends to keep wavefunctions with $l \neq 0$ away from the origin. Dividing through this equation by $E = -|E|$ (a negative quantity since $E \leqslant 0$ for a bound state) and making the substitution

$$\rho^2 = \frac{2m_e|E|r^2}{\hbar^2} \tag{2.17}$$

reduces the equation to the dimensionless form

$$\frac{d^2P}{d\rho^2} + \left\{-\frac{l(l+1)}{\rho^2} + \frac{\lambda}{\rho} - 1\right\}P = 0\,. \tag{2.18}$$

The constant that characterises the Coulomb interaction strength is

$$\lambda = \frac{e^2}{4\pi\epsilon_0}\sqrt{\frac{2m_e}{\hbar^2|E|}}\,. \tag{2.19}$$

The standard method of solving such differential equations is to look for a solution in the form of a series. The series solutions have a finite number of terms and do not diverge when $\lambda = 2n$, where n is an integer.[15] Thus, from eqn 2.19, these wavefunctions have eigenenergies given by[16]

$$E = -\frac{2m_e\left(e^2/4\pi\epsilon_0\right)^2}{\hbar^2}\frac{1}{\lambda^2} = -hcR_\infty\frac{1}{n^2}\,. \tag{2.20}$$

This shows that the Schrödinger equation has stationary solutions at energies given by the Bohr formula. The energy does not depend on l; this *accidental* degeneracy of wavefunctions with different l is a special feature of Coulomb potential. In contrast, degeneracy with respect to the magnetic quantum number m_l arises because of the system's symmetry, i.e. an atom's properties are independent of its orientation in space, in the absence of external fields.[17] The solution of the Schrödinger equation gives much more information than just the energies; from the wavefunctions we can calculate other atomic properties in ways that were not possible in the Bohr–Sommerfeld theory.

We have not gone through the gory details of the series solution, but we should examine a few examples of radial wavefunctions (see Table 2.2). Although the energy depends only on n, the shape of the wavefunctions depends on both n and l and these two quantum numbers are used to label the radial functions $R_{n,l}(r)$. For $n = 1$ there is only the $l = 0$ solution, namely $R_{1,0} \propto e^{-\rho}$. For $n = 2$ the orbital angular momentum quantum number is $l = 0$ or 1, giving

$$R_{2,0} \propto (1-\rho)\,e^{-\rho}\,,$$
$$R_{2,1} \propto \rho e^{-\rho}\,.$$

[15] The solution has the general form $P(\rho) = Ce^{-\rho}v(\rho)$, where $v(\rho)$ is another function of the radial coordinate, for which there is a polynomial solution (see Woodgate 1980 and Rae 1992).

[16] Using eqn 1.41.

[17] This is true for any spherically-symmetric potential $V(r)$.

Table 2.2 Radial hydrogenic wavefunctions $R_{n,l}$ in terms of the variable $\rho = Zr/(na_0)$, which gives a scaling that varies with n. The Bohr radius a_0 is defined in eqn 1.40.

$$R_{1,0} = \left(\frac{Z}{a_0}\right)^{3/2} 2\,\mathrm{e}^{-\rho}$$

$$R_{2,0} = \left(\frac{Z}{2a_0}\right)^{3/2} 2\,(1-\rho)\,\mathrm{e}^{-\rho}$$

$$R_{2,1} = \left(\frac{Z}{2a_0}\right)^{3/2} \frac{2}{\sqrt{3}}\,\rho\,\mathrm{e}^{-\rho}$$

$$R_{3,0} = \left(\frac{Z}{3a_0}\right)^{3/2} 2\left(1-2\rho+\frac{2}{3}\rho^2\right)\mathrm{e}^{-\rho}$$

$$R_{3,1} = \left(\frac{Z}{3a_0}\right)^{3/2} \frac{4\sqrt{2}}{3}\,\rho\left(1-\frac{1}{2}\rho\right)\mathrm{e}^{-\rho}$$

$$R_{3,2} = \left(\frac{Z}{3a_0}\right)^{3/2} \frac{2\sqrt{2}}{3\sqrt{5}}\,\rho^2\,\mathrm{e}^{-\rho}$$

$$\text{Normalisation: } \int_0^\infty R_{n,l}^2\, r^2\, \mathrm{d}r = 1$$

These show a general a feature of hydrogenic wavefunctions, namely that the radial functions for $l = 0$ have a finite value at the origin, i.e. the power series in ρ starts at the zeroth power. Thus electrons with $l = 0$ (called s-electrons) have a finite probability of being found at the position of the nucleus and this has important consequences in atomic physics.

Inserting $|E|$ from eqn 2.20 into eqn 2.17 gives the scaled coordinate

$$\rho = \frac{Z}{n}\frac{r}{a_0}\,, \tag{2.21}$$

where the atomic number has been incorporated by the replacement $e^2/4\pi\epsilon_0 \rightarrow Ze^2/4\pi\epsilon_0$ (as in Chapter 1). There are some important properties of the radial wavefunctions that require a general form of the solution and for future reference we state these results. The probability density of electrons with $l = 0$ at the origin is

$$|\psi_{n,l=0}(0)|^2 = \frac{1}{\pi}\left(\frac{Z}{na_0}\right)^3 . \tag{2.22}$$

For electrons with $l \neq 0$ the expectation value of $1/r^3$ is

$$\left\langle \frac{1}{r^3} \right\rangle = \int_0^\infty \frac{1}{r^3} R_{n,l}^2(r)\, r^2\, \mathrm{d}r = \frac{1}{l\left(l+\frac{1}{2}\right)(l+1)}\left(\frac{Z}{na_0}\right)^3 . \tag{2.23}$$

These results have been written in a form that is easy to remember; they must both depend on $1/a_0^3$ in order to have the correct dimensions and the dependence on Z follows from the scaling of the Schrödinger equation. The dependence on the principal quantum number n also seems to follow from eqn 2.21 but this is coincidental; a counterexample is[18]

$$\left\langle \frac{1}{r} \right\rangle = \frac{1}{n^2}\left(\frac{Z}{a_0}\right). \tag{2.24}$$

[18] This quantity is related to the quantum mechanical expectation value of the potential energy $\langle \text{p.e.} \rangle$; as in the Bohr model the total energy is $E = \langle \text{p.e.} \rangle / 2$.

2.2 Transitions

The wavefunction solutions of the Schrödinger equation for particular energies are standing waves and give a distribution of electronic charge $-e\left|\psi\left(r\right)\right|^2$ that is constant in time. We shall now consider how transitions between these stationary states occur when the atom interacts with electromagnetic radiation that produces an oscillating electric field[19]

$$\mathbf{E}\left(t\right) = \left|\mathbf{E}_0\right| \mathrm{Re}\left(\mathrm{e}^{-\mathrm{i}\omega t}\,\widehat{\mathbf{e}}_{\mathrm{rad}}\right) \tag{2.25}$$

with constant amplitude $\left|\mathbf{E}_0\right|$ and polarization vector $\widehat{\mathbf{e}}_{\mathrm{rad}}$.[20] If ω lies close to the atomic resonance frequency then the perturbing electric field puts the atom into a superposition of different states and induces an oscillating electric dipole moment on the atom (see Exercise 2.10). The calculation of the stimulated transition rate requires time-dependent perturbation theory (TDPT), as described in Chapter 7. However, the treatment from first principles is lengthy and we shall anticipate some of the results so that we can see how spectra relate to the underlying structure of the atomic energy levels. This does not require an exact calculation of transition rates, but we only need to determine whether the transition rate has a finite value or whether it is zero (to first order), i.e. whether the transition is allowed and gives a strong spectral line, or is forbidden.

The result of time-dependent perturbation theory is encapsulated in the golden rule (or Fermi's golden rule);[21] this states that the rate of transitions is proportional to the square of the matrix element of the perturbation. The Hamiltonian that describes the time-dependent interaction with the field in eqn 2.25 is $H' = e\mathbf{r}\cdot\mathbf{E}\left(t\right)$, where the electric dipole operator is $-e\mathbf{r}$.[22] This interaction with the radiation stimulates transitions from state 1 to state 2 at a rate[23]

$$\text{Rate} \propto \left|e\mathbf{E}_0\right|^2 \left|\int \psi_2^*\left(\mathbf{r}\cdot\widehat{\mathbf{e}}_{\mathrm{rad}}\right)\psi_1\,\mathrm{d}^3\mathbf{r}\right|^2 \equiv \left|e\mathbf{E}_0\right|^2 \times \left|\langle 2|\,\mathbf{r}\cdot\widehat{\mathbf{e}}_{\mathrm{rad}}\,|1\rangle\right|^2 . \tag{2.26}$$

The concise expression in Dirac notation is convenient for later use. This treatment assumes that the amplitude of the electric field is uniform over the atom so that it can be taken outside the integral over the atomic wavefunctions, i.e. that $\mathbf{E}_0$ does not depend on $\mathbf{r}$.[24] We write the dipole matrix element as the product

$$\langle 2|\,\mathbf{r}\cdot\widehat{\mathbf{e}}_{\mathrm{rad}}\,|1\rangle = D_{12}\,\mathcal{I}_{\mathrm{ang}}\,. \tag{2.27}$$

The radial integral is[25]

[19] The interaction of atoms with the oscillating magnetic field in such a wave is considerably weaker; see Appendix C.

[20] The unit vector $\widehat{\mathbf{e}}_{\mathrm{rad}}$ gives the direction of the oscillating electric field. For example, for the simple case of linear polarization along the x-axis $\widehat{\mathbf{e}}_{\mathrm{rad}} = \widehat{\mathbf{e}}_x$ and the real part of $\mathrm{e}^{-\mathrm{i}\omega t}$ is $\cos(\omega t)$; therefore $\mathbf{E}\left(t\right) = \left|\mathbf{E}_0\right|\cos(\omega t)\,\widehat{\mathbf{e}}_x$.

[21] See quantum mechanics texts such as Mandl (1992).

[22] This is analogous to the interaction of a classical dipole with an electric field. Atoms do not have a permanent dipole moment, but one is induced by the oscillating electric field. For a more rigorous derivation, see Woodgate (1980) or Loudon (2000).

[23] The maximum transition rate occurs when ω, the frequency of the radiation, matches the transition frequency ω_{12}, as discussed in Chapter 7. Note, however, that we shall not discuss the so-called 'density of states' in the golden rule since this is not straightforward for monochromatic radiation.

[24] In eqn 2.25 the phase of the wave is actually $(\omega t - \mathbf{k}\cdot\mathbf{r})$, where $\mathbf{r}$ is the coordinate relative to the atom's centre of mass (taken to be the origin) and $\mathbf{k}$ is the wavevector. We assume that the variation of phase $\mathbf{k}\cdot\mathbf{r}$ is small over the atom ($ka_0 \ll 2\pi$). This is equivalent to $\lambda \gg a_0$, i.e. the radiation has a wavelength much greater than the size of the atom. This is called the dipole approximation.

[25] Note that $D_{12} = D_{21}$.

$$D_{12} = \int_0^\infty R_{n_2,l_2}(r)\; r\, R_{n_1,l_1}(r)\, r^2\, \mathrm{d}r\,. \tag{2.28}$$

The angular integral is

$$\mathcal{I}_{\mathrm{ang}} = \int_0^{2\pi}\int_0^{\pi} Y^*_{l_2,m_2}(\theta,\phi)\; \widehat{\mathbf{r}}\cdot\widehat{\mathbf{e}}_{\mathrm{rad}}\, Y_{l_1,m_1}(\theta,\phi)\, \sin\theta\, \mathrm{d}\theta\, \mathrm{d}\phi\,, \tag{2.29}$$

where $\widehat{\mathbf{r}} = \mathbf{r}/r$. The radial integral is not normally zero although it can be small for transitions between states whose radial wavefunctions have a small overlap, e.g. when n_1 is small and n_2 is large (or the other way round). In contrast, the $\mathcal{I}_{\mathrm{ang}} = 0$ unless strict criteria are satisfied—these are the selection rules.

2.2.1 Selection rules

The selection rules that govern allowed transitions arise from the angular integral in eqn 2.29 which contains the angular dependence of the interaction $\widehat{\mathbf{r}}\cdot\widehat{\mathbf{e}}_{\mathrm{rad}}$ for a given polarization of the radiation. The mathematics requires that we calculate $\mathcal{I}_{\mathrm{ang}}$ for an atom with a well-defined quantisation axis (invariably chosen to be the z-axis) and radiation that has a well-defined polarization and direction of propagation. This corresponds to the physical situation of an atom experiencing the Zeeman effect of an external magnetic field, as described in Section 1.8; that treatment of the electron as a classical oscillator showed that the components of different frequencies within the Zeeman pattern have different polarizations. We use the same nomenclature of π- and σ-transitions here; transverse observation refers to radiation emitted perpendicular to the magnetic field, and longitudinal observation is along the z-axis.[26]

[26]If either the atoms have random orientations (e.g. because there is no external field) or the radiation is unpolarized (or both), then an average over all angles must be made at the end of the calculation.

To calculate $\mathcal{I}_{\mathrm{ang}}$ we write the unit vector $\widehat{\mathbf{r}}$ in the direction of the induced dipole as:

$$\begin{aligned}\widehat{\mathbf{r}} &= \frac{1}{r}\left(x\widehat{\mathbf{e}}_x + y\widehat{\mathbf{e}}_y + z\widehat{\mathbf{e}}_z\right)\\ &= \sin\theta\cos\phi\,\widehat{\mathbf{e}}_x + \sin\theta\sin\phi\,\widehat{\mathbf{e}}_y + \cos\theta\,\widehat{\mathbf{e}}_z\,.\end{aligned} \tag{2.30}$$

Expressing the functions of θ and ϕ in terms of spherical harmonic functions as

$$\begin{aligned}\sin\theta\cos\phi &= \sqrt{\frac{2\pi}{3}}\left(Y_{1,-1} - Y_{1,1}\right),\\ \sin\theta\sin\phi &= \mathrm{i}\sqrt{\frac{2\pi}{3}}\left(Y_{1,-1} + Y_{1,1}\right),\\ \cos\theta &= \sqrt{\frac{4\pi}{3}}Y_{1,0}\,,\end{aligned} \tag{2.31}$$

leads to

$$\widehat{\mathbf{r}} \propto Y_{1,-1}\frac{\widehat{\mathbf{e}}_x + \mathrm{i}\widehat{\mathbf{e}}_y}{\sqrt{2}} + Y_{1,0}\widehat{\mathbf{e}}_z + Y_{1,1}\frac{-\widehat{\mathbf{e}}_x + \mathrm{i}\widehat{\mathbf{e}}_y}{\sqrt{2}}\,. \tag{2.32}$$

We write the general polarization vector as

$$\widehat{\mathbf{e}}_{\mathrm{rad}} = A_{\sigma^-}\frac{\widehat{\mathbf{e}}_x - \mathrm{i}\widehat{\mathbf{e}}_y}{\sqrt{2}} + A_\pi\widehat{\mathbf{e}}_z + A_{\sigma^+}\left(-\frac{\widehat{\mathbf{e}}_x + \mathrm{i}\widehat{\mathbf{e}}_y}{\sqrt{2}}\right), \tag{2.33}$$

where A_π depends on the component of the electric field along the z-axis and the component in the xy-plane is written as a superposition of two circular polarizations with amplitudes A_{σ^+} and A_{σ^-} (rather than in terms of linear polarization in a Cartesian basis).[27] Similarly, the classical motion of the electron was written in terms of three eigenvectors in Section 1.8: an oscillation along the z-axis and circular motion in the xy-plane, both clockwise and anticlockwise.

From the expression for $\widehat{\mathbf{r}}$ in terms of the angular functions $Y_{l,m}(\theta,\phi)$ with $l=1$ we find that the dipole induced on the atom is proportional to[28]

$$\widehat{\mathbf{r}} \cdot \widehat{\mathbf{e}}_{\mathrm{rad}} \propto A_{\sigma^-} Y_{1,-1} + A_z Y_{1,0} + A_{\sigma^+} Y_{1,+1} \,. \tag{2.34}$$

The following sections consider the transitions that arise from these three terms.[29]

π-transitions

The component of the electric field along the z-axis A_z induces a dipole moment on the atom proportional to $\widehat{\mathbf{e}}_{\mathrm{rad}} \cdot \widehat{\mathbf{e}}_z = \cos\theta$ and the integral over the angular parts of the wavefunctions is

$$\mathcal{I}^{\pi}_{\mathrm{ang}} = \int_0^{2\pi}\int_0^{\pi} Y^*_{l_2,m_2}(\theta,\phi)\, \cos\theta\, Y_{l_1,m_1}(\theta,\phi)\, \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi \,. \tag{2.35}$$

To determine this integral we exploit the symmetry with respect to rotations about the z-axis.[30] The system has cylindrical symmetry, so the value of this integral is unchanged by a rotation about the z-axis through an angle ϕ_0:

$$\mathcal{I}^{\pi}_{\mathrm{ang}} = \mathrm{e}^{\mathrm{i}(m_1 - m_2)\phi_0} \mathcal{I}^{\pi}_{\mathrm{ang}} \,. \tag{2.36}$$

This equation is satisfied if either $\mathcal{I}^{\pi}_{\mathrm{ang}} = 0$ or $m_{l_1} = m_{l_2}$. For this polarization the magnetic quantum number does not change, $\Delta m_l = 0$.[31]

σ-transitions

The component of the oscillating electric field in the xy-plane excites σ-transitions. Equation 2.34 shows that the circularly-polarized radiation with amplitude A_{σ^+} excites an oscillating dipole moment on the atom proportional $Y_{1,1} \propto \sin\theta\, \mathrm{e}^{\mathrm{i}\phi}$, for which the angular integral is

$$\mathcal{I}^{\sigma^+}_{\mathrm{ang}} = \int_0^{2\pi}\int_0^{\pi} Y^*_{l_2,m_2}(\theta,\phi)\, \sin\theta\, \mathrm{e}^{\mathrm{i}\phi}\, Y_{l_1,m_1}(\theta,\phi)\, \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi \,. \tag{2.37}$$

Again, consideration of symmetry with respect to rotation about the z-axis through an arbitrary angle shows that $\mathcal{I}^{\sigma^+}_{\mathrm{ang}} = 0$ unless $m_{l_1} - m_{l_2} + 1 = 0$. The interaction of an atom with circularly-polarized radiation of the opposite handedness leads to a similar integral but with $\mathrm{e}^{\mathrm{i}\phi} \rightarrow \mathrm{e}^{-\mathrm{i}\phi}$; this integral $\mathcal{I}^{\sigma^-}_{\mathrm{ang}} = 0$ unless $m_{l_1} - m_{l_2} - 1 = 0$. Thus the selection rule for the σ-transitions is $\Delta m_l = \pm 1$.

We have found the selection rules that govern Δm_l for each of the three possible polarizations of the radiation separately. These apply

[27] We will see that the labels π, σ^+ and σ^- refer to the transition that the radiation excites; for this it is only important to know how the electric field behaves at the position of the atom. The polarization state associated with this electric field, e.g. whether it is right- or left-handed circularly-polarized radiation, also depends on the direction of propagation (wavevector), but we shall try to avoid a detailed treatment of the polarization conventions in this discussion of the principles. Clearly, however, it is important to have the correct polarization when setting up actual experiments.

[28] The eigenvectors have the following properties:

$$\frac{\widehat{\mathbf{e}}_x + \mathrm{i}\widehat{\mathbf{e}}_y}{\sqrt{2}} \cdot \frac{\widehat{\mathbf{e}}_x - \mathrm{i}\widehat{\mathbf{e}}_y}{\sqrt{2}} = 1$$

and

$$\frac{\widehat{\mathbf{e}}_x \pm \mathrm{i}\widehat{\mathbf{e}}_y}{\sqrt{2}} \cdot \frac{\widehat{\mathbf{e}}_x \pm \mathrm{i}\widehat{\mathbf{e}}_y}{\sqrt{2}} = 0 \,.$$

[29] In spherical tensor notation (Woodgate 1980) the three vector components are written A_{-1}, A_0 and A_{+1}, which is convenient for more general use; but writing eqn 2.34 as given emphasises that the amplitudes A represent the different polarizations of the radiation and the spherical harmonics come from the atomic response (induced dipole moment).

[30] Alternative methods are given below and in Exercise 2.9.

[31] We use m_l to distinguish this quantum number from m_s, the magnetic quantum number for spin angular momentum that is introduced later. Specific functions of the spatial variables such as $Y_{l,m}$ and $\mathrm{e}^{-\mathrm{i}m\phi}$ do not need this additional subscript.

when the polarized light interacts with an atom that has a well-defined orientation, e.g. an atom in an external magnetic field. If the light is unpolarized or there is no defined quantisation axis, or both, then $\Delta m_l = 0, \pm 1$.

Example 2.1 *Longitudinal observation*
Electromagnetic radiation is a transverse wave with its oscillating electric field perpendicular to the direction of propagation, $\widehat{\mathbf{e}}_{\mathrm{rad}} \cdot \mathbf{k} = 0$. Thus radiation with wavevector $\mathbf{k} = k\widehat{\mathbf{e}}_z$ has $A_z = 0$ and π-transitions do not occur.[32] Circularly-polarized radiation (propagating along the z-axis) is a special case for which transitions occur with either $\Delta m_l = +1$ or $\Delta m_l = -1$, depending on the handedness of the radiation, but not both.

[32]Similar behaviour arises in the classical model of the normal Zeeman effect in Section 1.8, but the quantum treatment in this section shows that it is a general feature of longitudinal observation—not just for the normal Zeeman effect.

2.2.2 Integration with respect to θ

In the angular integral the spherical harmonic functions with $l = 1$ (from eqn 2.34) are sandwiched between the angular momentum wavefunctions of the initial and final states so that

$$\mathcal{I}_{\mathrm{ang}} \propto \int_0^{2\pi}\int_0^{\pi} Y^*_{l_2,m_2}\, Y_{1,m}\, Y_{l_1,m_1} \sin\theta\, \mathrm{d}\theta\, \mathrm{d}\phi\,. \tag{2.38}$$

To calculate this angular integral we use the following formula:[33]

$$Y_{1,m}\, Y_{l_1,m_1} = A\, Y_{l_1+1,\, m_1+m} + B\, Y_{l_1-1, m_1+m}\,, \tag{2.39}$$

where A and B are constants whose exact values need not concern us. Thus from the orthogonality of the spherical harmonics[34] we find

$$\mathcal{I}_{\mathrm{ang}} \propto A\, \delta_{l_2,l_1+1}\delta_{m_2,m_1+m} + B\, \delta_{l_2,l_1-1}\delta_{m_2,m_1+m}.$$

[33]See the references on angular momentum in quantum mechanics; the reason why the magnetic quantum numbers add is obvious from $\Phi(\phi)$.

[34]We have $\int_0^{2\pi}\int_0^{\pi} Y_{l',m'} Y_{l,m} \sin\theta\, \mathrm{d}\theta\, \mathrm{d}\phi = \delta_{l',l}\, \delta_{m',m}$. This reduces to the normalisation in Table 2.1 when $l' = l$ and $m' = m$.

The delta functions give the selection rule found previously, namely $\Delta m_l = m$, where $m = 0, \pm 1$ depending on the polarization, and also $\Delta l = \pm 1$. In the mathematics the functions with $l = 1$ that represent the interaction with the radiation are sandwiched between the orbital angular momentum eigenfunctions of the initial and final states. Thus the rule $\Delta l = \pm 1$ can be interpreted as conservation of angular momentum for a photon carrying one unit of angular momentum, $\hbar$ (Fig. 2.8 illustrates this reasoning for the case of total angular momentum).[35] The changes in the magnetic quantum number are also consistent with this picture—the component of the photon's angular momentum along the z-axis being $\Delta m_l = 0, \pm 1$. Conservation of angular momentum does not explain why $\Delta l \neq 0$—this comes about because of parity, as explained below.

[35]This argument applies only for electric dipole radiation. Higher-order terms, e.g. quadrupole radiation, can give $\Delta l > 1$.

2.2.3 Parity

Parity is an important symmetry property throughout atomic and molecular physics and its general use will be explained before applying it to selection rules. The parity transformation is an inversion through the origin given by $\mathbf{r} \to -\mathbf{r}$. This is equivalent to the following transformation of the polar coordinates:

$$\begin{aligned} \theta &\longrightarrow \pi - \theta : \quad \text{a reflection}\,, \\ \phi &\longrightarrow \phi + \pi : \quad \text{a rotation}\,. \end{aligned}$$

The reflection produces a mirror image of the original system and parity is also referred to as mirror symmetry. The mirror image of a hydrogen atom has the same energy levels as those in the original atom since the Coulomb potential is the same after reflection. It turns out that all the electric and magnetic interactions 'look the same' after reflection and all atoms have parity symmetry.[36] To find the eigenvalues for parity we use the full quantum mechanical notation, with hats to distinguish the operator $\widehat{P}$ from its eigenvalue P in the equation

$$\widehat{P}\,\psi = P\,\psi\,, \tag{2.40}$$

from which it follows that $\widehat{P}^2\psi = P^2\,\psi$. Two successive parity operations correspond to there being no change (the identity operator), i.e. $\mathbf{r} \to -\mathbf{r} \to \mathbf{r}$. Thus $P^2 = 1$. Therefore the parity operator has eigenvalues $P = 1$ and -1 that correspond to even and odd parity wavefunctions, respectively:

$$\widehat{P}\,\psi = \psi \qquad \text{or} \qquad \widehat{P}\,\psi = -\,\psi\,.$$

Both eigenvalues occur for the spherical harmonic functions,

$$\widehat{P}\;Y_{l,m} = (-1)^{\,l}\,Y_{l,m}\,. \tag{2.41}$$

The value of the angular integral does not change in a parity transformation[37] so

$$\mathcal{I}_{\text{ang}} = (-1)^{\,l_2+l_1+1}\,\mathcal{I}_{\text{ang}}\,. \tag{2.42}$$

Thus the integral is zero unless the initial and final states have opposite parity (see Exercise 2.12). In particular, electric dipole transitions require an odd change in the orbital angular momentum quantum number ($\Delta l \neq 0$).[38]

The treatment above of the parity operator acting on a wavefunction is quite general and even in complex atoms the wavefunctions have a definite parity. The selection rules we have discussed in this section and others are tabulated in Appendix C. If the electric dipole matrix element is zero between two states then other types of transition may occur but at a rate many orders of magnitude slower than allowed transitions.

The allowed transitions between the $n = 1$, 2 and 3 shells of atomic hydrogen are shown in Fig. 2.2, as an example of the selection rules. The 2s configuration has no allowed transitions downwards; this makes it metastable, i.e. it has a very long lifetime of about 0.125 s.[39]

Finally, a comment on the spectroscopic notation. It can be seen in Fig. 2.2 that the allowed transitions give rise to several series of

[36] This can be proved formally in quantum mechanics by showing that the Hamiltonians for these interactions commute with the parity operator. The weak interaction in nuclear physics does not have mirror symmetry and violates parity conservation. The extremely small effect of the weak interaction on atoms has been measured in exceedingly careful and precise experiments.

[37] See, for example, Mandl (1992).

[38] The radial integral is not changed by the parity transformation.

[39] This special feature is used in the experiment described in Section 2.3.4.

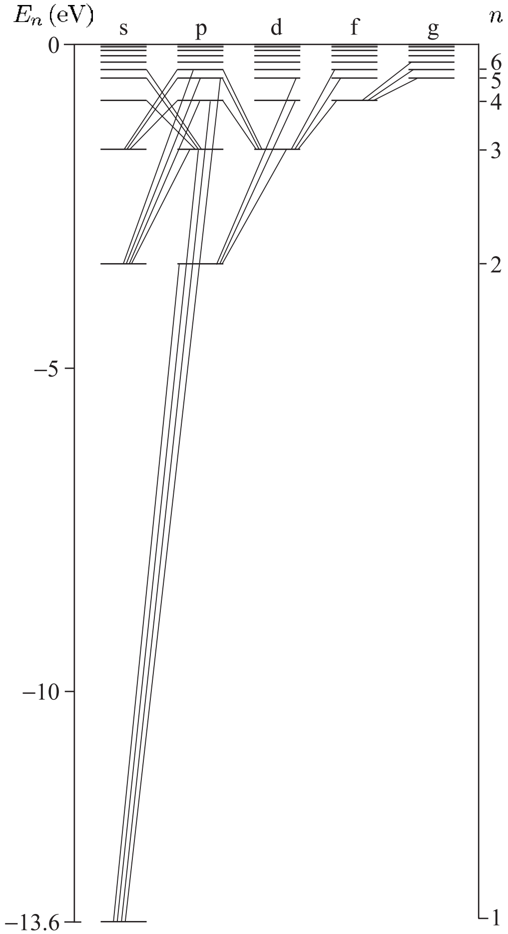

Fig. 2.2 Allowed transitions between the configurations of hydrogen obey the selection rule $\Delta l = \pm 1$. The configurations with $l = 0, 1, 2, 3, 4, \ldots$ are labelled s, p, d, f, g, and so on alphabetically (the usual convention). In the special case of hydrogen the energy does not depend on the quantum number l.

lines. The series of lines to the ground configuration is called the p-series, where p stands for principal—this is the only series observed in absorption[40]—hence p labels configurations with $l = 1$. The s-series of lines goes from $l = 0$ configurations (to a level with $l = 1$), and similarly the d-series goes from $l = 2$ configurations; s and d stand for sharp and diffuse, respectively.[41]

[40]For hydrogen this is the Lyman series, as marked on Fig. 1.1; however, p-series is a general name.

[41]These names reflect the appearance of the lines in the first experimental observations.

2.3 Fine structure

Relativistic effects lead to small splittings of the atomic energy levels called fine structure. We estimated the size of this structure in Section 1.4 by comparing the speed of electrons in classical orbits with the speed of light.[42] In this section we look at how to calculate fine structure by treating relativistic effects as a perturbation to the solutions of the Schrödinger equation. This approach requires the concept that electrons have spin.

[42]By considering elliptical orbits, rather than just circular ones, Sommerfeld refined Bohr's theory to obtain a relativistic expression for the energy levels in hydrogen that gave very accurate predictions of the fine structure; however, details of that approach are not given here.

2.3.1 Spin of the electron

In addition to the evidence provided by observations of the fine structure itself, that is described in this section, two other experiments showed that the electron has spin angular momentum, not just orbital angular momentum. One of these pieces of experimental evidence for spin was the observation of the so-called anomalous Zeeman effect. For many atoms, e.g. hydrogen and sodium, the splitting of their spectral lines in a magnetic field does not have the pattern predicted by the normal Zeeman effect (that we found classically in Section 1.8). This anomalous Zeeman effect has a straightforward explanation in terms of electron spin (as shown in Section 5.5). The second experiment was the famous Stern–Gerlach experiment that will be described in Section 6.4.1.[43]

Unlike orbital angular momentum, spin does not have eigenstates that are functions of the angular coordinates. Spin is a more abstract concept and it is convenient to write its eigenstates in Dirac's ket notation as $|s\, m_s\rangle$. The full wavefunction for a one-electron atom is the product of the radial, angular and spin wavefunctions: $\Psi = R_{n,l}(r)\, Y_{l,m}\,(\theta,\phi)\,|s\, m_s\rangle$. Or, using ket notation for all of the angular momentum, not just the spin,

$$\Psi = R_{n,l}(r)\ |l\, m_l\, s\, m_s\rangle\,. \tag{2.43}$$

These atomic wavefunctions provide a basis in which to calculate the effect of perturbations on the atom. However, some problems do not require the full machinery of (degenerate) perturbation theory and for the time being we shall treat the orbital and spin angular momenta by analogy with classical vectors. To a large extent this vector model is intuitively obvious and we start to use it without formal derivations. But note the following points. An often-used shorthand for the spin eigenfunctions is spin-up:

$$\left|s = \tfrac{1}{2},\ m_s = \tfrac{1}{2}\right\rangle \equiv |\uparrow\rangle\,, \tag{2.44}$$

and similarly $|\downarrow\rangle$ for the $m_s = -\frac{1}{2}$ state (spin-down). However, in quantum mechanics the angular momentum cannot be completely aligned 'up' or 'down' with respect to the z-axis, otherwise the x- and y-components would be zero and we would know all three components simultaneously.[44] The vector model mimics this feature with classical vectors drawn with length $|\mathbf{s}| = \sqrt{s(s+1)} = \sqrt{3}/2$. (Only the expectation value of the square of the angular momentum has meaning in quantum mechanics.) The spin-up and spin-down states are as illustrated in Fig. 2.3 with components along the z-axis of $\pm\frac{1}{2}$. We can think of the vector as rotating around the z-axis, or just having an undefined direction in the xy-plane corresponding to a lack of knowledge of the x- and y-components (see also Grant and Phillips 2001).

The name 'spin' invokes an analogy with a classical system spinning on its axis, e.g. a sphere rotating about an axis through its centre of mass, but this mental picture has to be treated with caution; spin cannot be equal to the sum of the orbital angular momenta of the constituents since that will always be an integer multiple of $\hbar$. In any case, the electron is

[43] The fine structure, anomalous Zeeman effect and Stern–Gerlach experiment all involve the interaction of the electron's magnetic moment with a magnetic field—the internal field of the atom in the case of fine structure. Stern and Gerlach detected the magnetic interaction by its influence on the atom's motion, whereas the Zeeman effect and fine structure are observed by spectroscopy.

[44] This is not possible since the operators for the x-, y- and z-components of angular momentum do not commute (save in a few special cases; we can know that $s_x = s_y = s_z = 0$ if $s = 0$).

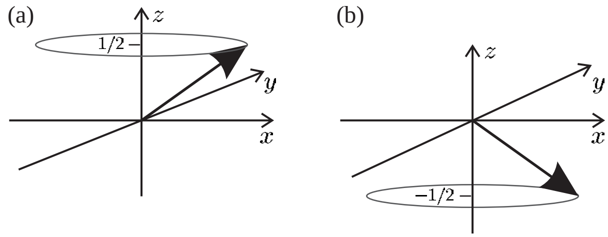

Fig. 2.3 The representation of (a) spin-up and (b) spin-down states as vectors precessing around the z-axis.

a structureless elementary particle with no measurable size. So we are left with the experimental fact that the electron has an intrinsic spin angular momentum of $\hbar/2$ and these half-integer values are perfectly acceptable within the general theory of angular momentum in quantum mechanics.

2.3.2 The spin–orbit interaction

The Schrödinger equation is non-relativistic, as can readily be seen by looking at the kinetic-energy operator that is equivalent to the non-relativistic expression $p^2/2m_e$. Some of the relativistic effects can be taken into account as follows. An electron moving through an electric field $\mathbf{E}$ experiences an effective magnetic field $\mathbf{B}$ given by

$$\mathbf{B} = -\frac{1}{c^2}\mathbf{v} \times \mathbf{E}\,. \tag{2.45}$$

This is a consequence of the way an electric field behaves under a Lorentz transformation from a stationary to a moving frame in special relativity. Although a derivation of this equation is not given here, it is certainly plausible since special relativity and electromagnetism are intimately linked through the speed of light $c = 1/\sqrt{\epsilon_0\mu_0}$. This equation for the speed of electromagnetic waves in a vacuum comes from Maxwell's equations; ϵ_0 being associated with the electric field and μ_0 with the magnetic field. Rearrangement to give $\mu_0 = 1/\left(\epsilon_0 c^2\right)$ suggests that magnetic fields arise from electrodynamics and relativity.[45]

[45] The Biot–Savart law for the magnetic field from a current flowing along a straight wire can be recovered from the Lorentz transformation and Coulomb's law (Griffiths 1999). However, this link can only be made in this direction for simple cases and generally the phenomenon of magnetism cannot be 'derived' in this way.

We now manipulate eqn 2.45 into a convenient form, by substituting for the electric field in terms of the gradient of the potential energy V and unit vector in the radial direction:

$$\mathbf{E} = \frac{1}{e}\frac{\partial V}{\partial r}\frac{\mathbf{r}}{r}\,. \tag{2.46}$$

The factor of e comes in because the electron's potential energy V equals its charge $-e$ times the electrostatic potential. From eqn 2.45 we have

$$\mathbf{B} = \frac{1}{m_e c^2}\left(\frac{1}{er}\frac{\partial V}{\partial r}\right)\mathbf{r} \times m_e\mathbf{v} = \frac{\hbar}{m_e c^2}\left(\frac{1}{er}\frac{\partial V}{\partial r}\right)\mathbf{l}\,, \tag{2.47}$$

where the orbital angular momentum is $\hbar\mathbf{l} = \mathbf{r} \times m_e\mathbf{v}$. The electron has an intrinsic magnetic moment $\boldsymbol{\mu} = -g_s\mu_B\mathbf{s}$, where the spin has a magnitude of $|\mathbf{s}| = s = 1/2$ (in units of $\hbar$) and $g_s \simeq 2$, so the moment has a magnitude close to one Bohr magneton ($\mu_B = e\hbar/2m_e$). The interaction of the electron's magnetic moment with the orbital field gives the Hamiltonian

$$\begin{aligned} H &= -\boldsymbol{\mu}\cdot\mathbf{B} \\ &= g_s\mu_B\mathbf{s}\cdot\frac{\hbar}{m_e c^2}\left(\frac{1}{er}\frac{\partial V}{\partial r}\right)\mathbf{l}. \end{aligned} \tag{2.48}$$

However, this expression gives energy splittings about twice as large as observed. The discrepancy comes from the Thomas precession—a relativistic effect that arises because we are calculating the magnetic field in a frame of reference that is not stationary but rotates as the electron moves about the nucleus. The effect is taken into account by replacing g_s with $g_s - 1 \simeq 1$.[46] Finally, we find the spin–orbit interaction, including the Thomas precession factor, is[47]

$$H_{s-o} = (g_s - 1)\frac{\hbar^2}{2m_e^2c^2}\left(\frac{1}{r}\frac{\partial V}{\partial r}\right)\mathbf{s}\cdot\mathbf{l}. \tag{2.49}$$

For the Coulomb potential in hydrogen we have

$$\frac{1}{r}\frac{\partial V}{\partial r} = \frac{e^2/4\pi\epsilon_0}{r^3}. \tag{2.50}$$

The expectation value of this Hamiltonian gives an energy change of[48]

$$E_{s-o} = \frac{\hbar^2}{2m_e^2c^2}\frac{e^2}{4\pi\epsilon_0}\left\langle\frac{1}{r^3}\right\rangle\langle\mathbf{s}\cdot\mathbf{l}\rangle. \tag{2.51}$$

The separation into a product of radial and angular expectation values follows from the separability of the wavefunction. The integral $\langle 1/r^3\rangle$ is given in eqn 2.23. However, we have not yet discussed how to deal with interactions that have the form of dot products of two angular momenta; let us start by defining the total angular momentum of the atom as the sum of its orbital and spin angular momenta,

$$\mathbf{j} = \mathbf{l} + \mathbf{s}. \tag{2.52}$$

This is a conserved quantity for a system without any external torque acting on it, e.g. an atom in a field-free region of space. This is true both in classical and quantum mechanics, but we concentrate on the classical explanation in this section. The spin–orbit interaction between $\mathbf{l}$ and $\mathbf{s}$ causes these vectors to change direction, and because their sum is constrained to be equal to $\mathbf{j}$ they move around as shown in Fig. 2.4.[49] Squaring and rearranging eqn 2.52, we find that $2\,\mathbf{s}\cdot\mathbf{l} = \mathbf{j}^2 - \mathbf{l}^2 - \mathbf{s}^2$. Hence we can find the expectation value in terms of the known values for $\langle\mathbf{j}^2\rangle$, $\langle\mathbf{l}^2\rangle$ and $\langle\mathbf{s}^2\rangle$ as

$$\langle\mathbf{s}\cdot\mathbf{l}\rangle = \frac{1}{2}\{j(j+1) - l(l+1) - s(s+1)\}. \tag{2.53}$$

Thus the spin–orbit interaction produces a shift in energy of

$$E_{s-o} = \frac{\beta}{2}\{j(j+1) - l(l+1) - s(s+1)\}, \tag{2.54}$$

[46] This is almost equivalent to using $g_s/2 \simeq 1$, but $g_s - 1$ is more accurate at the level of precision where the small deviation of g_s from 2 is important (Haar and Curtis 1987). For further discussion of Thomas precession see Cowan (1981), Eisberg and Resnick (1985) and Munoz (2001).

[47] We have derived this classically, e.g. by using $\hbar\mathbf{l} = \mathbf{r}\times m_e\mathbf{v}$. However, the same expression can be obtained from the fully relativistic Dirac equation for an electron in a Coulomb potential by making a low-velocity approximation, see Sakurai (1967). That quantum mechanical approach justifies treating $\mathbf{l}$ and $\mathbf{s}$ as operators.

[48] Using the approximation $g_s - 1 \simeq 1$.

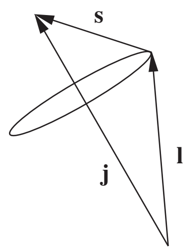

Fig. 2.4 The orbital and spin angular momenta add to give a total atomic angular momentum of $\mathbf{j}$.

[49] In this precession about $\mathbf{j}$ the magnitudes of $\mathbf{l}$ and $\mathbf{s}$ remain constant. The magnetic moment (proportional to $\mathbf{s}$) is not altered in an interaction with a magnetic field, and because of the symmetrical form of the interaction in eqn 2.49, we do not expect $\mathbf{l}$ to behave any differently. See also Blundell (2001) and Section 5.1.

where the spin–orbit constant β is (from eqns 2.51 and 2.23)

$$\beta = \frac{\hbar^2}{2m_e^2c^2}\,\frac{e^2}{4\pi\epsilon_0}\,\frac{1}{(na_0)^3\,l\left(l+\frac{1}{2}\right)(l+1)}\,. \tag{2.55}$$

A single electron has $s = \frac{1}{2}$ so, for each l, its total angular momentum quantum number j has two possible values:

$$j = l + \frac{1}{2} \quad \text{or} \quad l - \frac{1}{2}\,.$$

From eqn 2.54 we find that the energy interval between these levels, $\Delta E_{\mathrm{s-o}} = E_{j=l+\frac{1}{2}} - E_{j=l-\frac{1}{2}}$, is

$$\Delta E_{\mathrm{s-o}} = \beta\left(l+\tfrac{1}{2}\right) = \frac{\alpha^2 hcR_\infty}{n^3 l\,(l+1)}\,. \tag{2.56}$$

Or, expressed in terms of the gross energy $E(n)$ in eqn 1.10,[50]

$$\Delta E_{\mathrm{s-o}} = \frac{\alpha^2}{n\,l\,(l+1)}E\,(n)\,. \tag{2.57}$$

[50] As shown in Section 1.9, $m_e\alpha c a_0 = \hbar$ and $hcR_\infty = (e^2/4\pi\epsilon_0)/(2a_0)$.

This agrees with the qualitative discussion in Section 1.4, where we showed that relativistic effects cause energy changes of order α^2 times the gross structure. The more complete expression above shows that the energy intervals between levels decrease as n and l increase. The largest interval in hydrogen occurs for $n = 2$ and $l = 1$; for this configuration the spin–orbit interaction leads to levels with $j = 1/2$ and $j = 3/2$. The full designation of these levels is $2\mathrm{p}\,^2\mathrm{P}_{1/2}$ and $2\mathrm{p}\,^2\mathrm{P}_{3/2}$, in the notation that will be introduced for the LS-coupling scheme. But some of the quantum numbers (defined in Chapter 5) are superfluous for atoms with a single valence electron and a convenient short form is to denote these two levels by $2\,\mathrm{P}_{1/2}$ and $2\,\mathrm{P}_{3/2}$; these correspond to $n\,\mathrm{P}_j$, where P represents the (total) orbital angular momentum for this case. (The capital letters are consistent with later usage.) Similarly, we may write $2\,\mathrm{S}_{1/2}$ for the $2\mathrm{s}\,^2\mathrm{S}_{1/2}$ level; $3\,\mathrm{D}_{3/2}$ and $3\,\mathrm{D}_{5/2}$ for the $j = 3/2$ and $5/2$ levels, respectively, that arise from the 3d configuration.[51] But the full notation must be used whenever ambiguity might arise.

[51] Another short form found in the literature is $2\,^2\mathrm{P}_{1/2}$ and $2\,^2\mathrm{P}_{3/2}$.

2.3.3 The fine structure of hydrogen

As an example of fine structure, we look in detail at the levels that arise from the $n = 2$ and $n = 3$ shells of hydrogen. Equation 2.54 predicts that, for the 2p configuration, the fine-structure levels have energies of

$$\begin{aligned} E_{\mathrm{s-o}}\left(2\,\mathrm{P}_{1/2}\right) &= -\beta_{2\mathrm{p}}\,, \\ E_{\mathrm{s-o}}\left(2\,\mathrm{P}_{3/2}\right) &= \tfrac{1}{2}\beta_{2\mathrm{p}}\,, \end{aligned}$$

as shown in Fig. 2.5(a). For the 3d configuration

$$\begin{aligned} E_{\mathrm{s-o}}\left(3\,\mathrm{D}_{3/2}\right) &= -\frac{3}{2}\beta_{3\mathrm{d}}\,, \\ E_{\mathrm{s-o}}\left(3\,\mathrm{D}_{5/2}\right) &= \beta_{3\mathrm{d}}\,, \end{aligned}$$

as shown in Fig. 2.5(b). For both configurations, it is easy to see that the spin–orbit interaction does not shift the mean energy

$$\overline{E} = (2j+1)\,E_j(n,l) + (2j'+1)\,E_{j'}(n,l)\,, \qquad (2.58)$$

where $j' = l - 1/2$ and $j = l + 1/2$ for the two levels. This calculation of the 'centre of gravity' for all the states takes into account the degeneracy of each level.

The spin–orbit interaction does not affect the $2\,\mathrm{S}_{1/2}$ or $3\,\mathrm{S}_{1/2}$ so we might expect these levels to lie close to the centre of gravity of the configurations with $l > 0$. This is not the case. Fig. 2.6 shows the energies of the levels for the $n = 3$ shell given by a fully relativistic calculation. We can see that there are other effects of similar magnitude to the spin–orbit interaction that affect these levels in hydrogen. Quite remarkably, these additional relativistic effects shift the levels by just the right amount to make $n\,\mathrm{P}_{1/2}$ levels degenerate with the $n\,\mathrm{S}_{1/2}$ levels, and $n\,\mathrm{P}_{3/2}$ degenerate with $n\,\mathrm{D}_{3/2}$. This structure does not occur by chance, but points to a deeper underlying cause. The full explanation of these observations requires relativistic quantum mechanics and the technical details of such calculations lie beyond the scope of this book.[52] We shall simply quote the solution of the Dirac equation for an electron in a Coulomb potential; this gives a formula for the energy $E_{\mathrm{Dirac}}\,(n,j)$ that depends only on n and j, i.e. it gives the same energy for levels of the same n and j but different l, as in the cases above. In a comparison of the

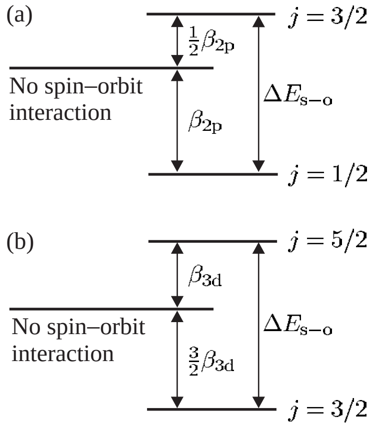

Fig. 2.5 The fine structure of hydrogen. The fine structure of (a) the 2p and (b) the 3d configurations are drawn on different scales: $\beta_{2\mathrm{p}}$ is considerably greater than $\beta_{3\mathrm{d}}$. All p- and d-configurations look similar apart from an overall scaling factor.

[52] See graduate-level quantum mechanics texts, e.g. Sakurai (1967) and Series (1988).

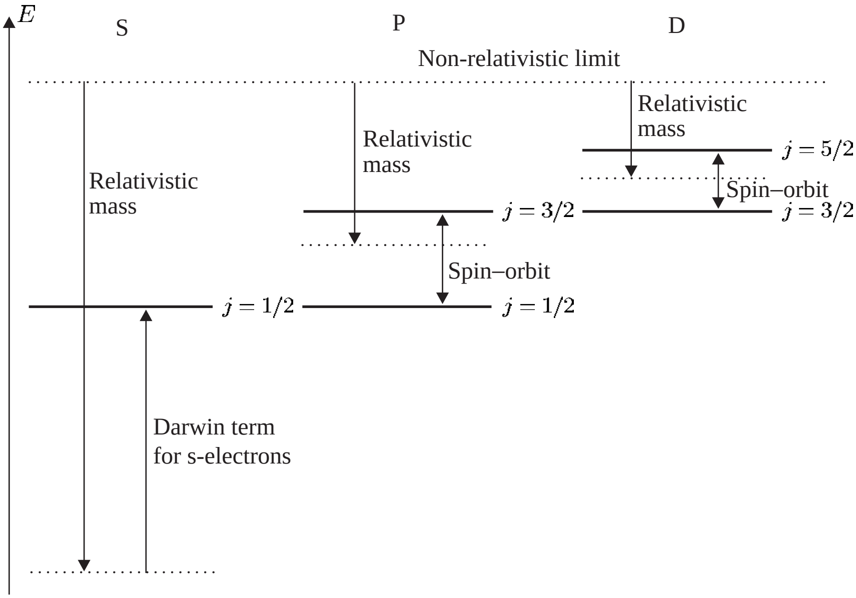

Fig. 2.6 The theoretical positions of the energy levels of hydrogen calculated by the fully relativistic theory of Dirac depend on n and j only (not l), as shown in this figure for the $n = 3$ shell. In addition to the spin–orbit interaction, the effects that determine the energies of these levels are: the relativistic mass correction and, for s-electrons only, the Darwin term (that accounts for relativistic effects that occur at small r, where the electron's momentum becomes comparable to $m_{\mathrm{e}}c$).

exact relativistic solution of the Dirac equation and the non-relativistic energy levels, three relativistic effects can be distinguished.

(a) There is a straightforward relativistic shift of the energy (or equivalently mass), related to the binomial expansion of $\gamma = (1-v^2/c^2)^{-1/2}$, in eqn 1.16. The term of order v^2/c^2 gives the non-relativistic kinetic energy $p^2/2m_e$. The next term in the expansion is proportional to v^4/c^4 and gives an energy shift of order v^2/c^2 times the gross structure—this is the effect that we estimated in Section 1.4.

(b) For electrons with $l \neq 0$, the comparison of the Dirac and Schrödinger equations shows that there is a spin–orbit interaction of the form given above, with the Thomas precession factor naturally included.[53]

(c) For electrons with $l = 0$ there is a **Darwin term** proportional to $|\psi(r=0)|^2$ that has no classical analogue (see Woodgate (1980) for further details).

That these different contributions conspire together to perturb the wavefunctions such that levels of the same n and j are degenerate seems improbable from a non-relativistic point of view. It is worth reiterating the statement above that this structure arises from the relativistic Dirac equation; making an approximation for small v^2/c^2 shows that these three corrections, and no others, need to be applied to the (non-relativistic) energies found from the Schrödinger equation.

2.3.4 The Lamb shift

Figure 2.7 shows the actual energy levels of the $n = 2$ and $n = 3$ shells. According to relativistic quantum theory the $2\,\mathrm{S}_{1/2}$ level should be exactly degenerate with $2\,\mathrm{P}_{1/2}$ because they both have $n = 2$ and $j = 1/2$, but in reality there is an energy interval between them, $E\left(2\,\mathrm{S}_{1/2}\right) - E\left(2\,\mathrm{P}_{1/2}\right) \simeq 1\,\mathrm{GHz}$. The shift of the $2\,\mathrm{S}_{1/2}$ level to a higher energy (lower binding energy) than the $E_{\mathrm{Dirac}}\,(n=2, j=1/2)$ is about one-tenth of the interval between the two fine-structure levels, $E\left(2\,\mathrm{P}_{3/2}\right) - E\left(2\,\mathrm{P}_{1/2}\right) \simeq 11\,\mathrm{GHz}$. Although small, this discrepancy in hydrogen was of great historical importance in physics. For this simple one-electron atom the predictions of the Dirac equation are very precise and that theory cannot account for Lamb and Retherford's experimental measurement that the $2\,\mathrm{S}_{1/2}$ level is indeed higher than the $2\,\mathrm{P}_{1/2}$ level.[54] The explanation of this **Lamb shift** goes beyond relativistic quantum mechanics and requires quantum electrodynamics (QED)—the quantum field theory that describes electromagnetic interactions. Indeed, the observation of the Lamb shift experiment was a stimulus for the development of this theory.[55] An intriguing feature of QED is so-called vacuum fluctuations—regions of free space are not regarded as being completely empty but are permeated by fluctuating electromagnetic fields.[56] The QED effects lead to a significant energy shift for electrons with $l = 0$ and hence break the degeneracy of $2\,\mathrm{S}_{1/2}$ and $2\,\mathrm{P}_{1/2}$.[57] The largest QED shift occurs for the $1\,\mathrm{S}_{1/2}$ ground level of hydrogen but there is no other level nearby and so a determination of its energy requires a precise mea-

[53] The Dirac equation predicts that the electron has $g_s = 2$ exactly.

[54] Lamb and Retherford used a radio-frequency to drive the $2\,\mathrm{S}_{1/2}$–$2\,\mathrm{P}_{1/2}$ transition directly. This small energy interval, now know as the Lamb shift, cannot be resolved in conventional spectroscopy because of Doppler broadening, but it can be seen using Doppler-free methods as shown in Fig. 8.7.

[55] The QED calculation of the Lamb shift is described in Sakurai (1967).

[56] Broadly speaking, in a mathematical treatment these vacuum fluctuations correspond to the zero-point energy of quantum harmonic oscillators, i.e. the lowest energy of the modes of the system is not zero but $\hbar\omega/2$.

[57] QED also explains why the g-factor of the electron is not exactly 2. Precise measurements show that $g_s = 2.002\,319\,304\,371\,8$ (current values for the fundamental constants can be found on the NIST web site, and those of other national standards laboratories). See also Chapter 12.

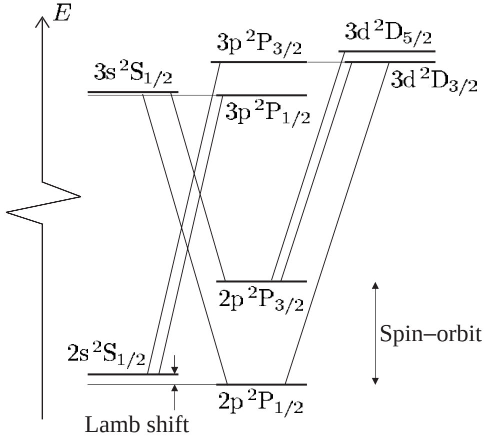

Fig. 2.7 The fine structure of the $n = 2$ and $n = 3$ shells of hydrogen and the allowed transitions between the levels. According to the Dirac equation, the $2\,\mathrm{S}_{1/2}$ and $2\,\mathrm{P}_{1/2}$ levels should be degenerate, but they are not. The measured positions show that the $2\mathrm{s}\,^2\mathrm{S}_{1/2}$ level is shifted upwards relative to the position $E_{\mathrm{Dirac}}\,(n = 2,\ j = 1/2)$ and is therefore not degenerate with the $2\mathrm{p}\,^2\mathrm{P}_{1/2}$ level. Such a shift occurs for all the s-electrons (but the size of the energy shift decreases with increasing n). The explanation of this shift takes us beyond relativistic quantum mechanics into the realm of quantum electrodynamics (QED)—the quantum field theory that describes electromagnetic interactions.

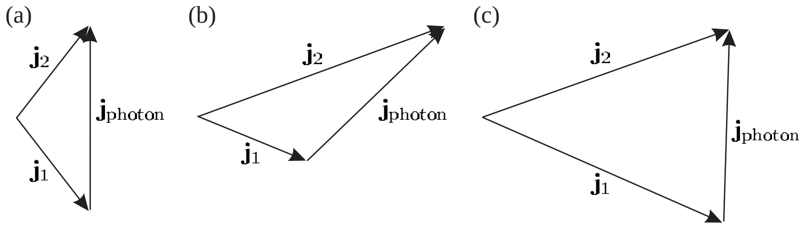

Fig. 2.8 The conservation of total angular momentum in electric dipole transitions that gives the selection rule in eqn 2.59 can be represented as vector addition. The photon has one unit of angular momentum, and so to go from level j_1 to j_2 the vectors must form a triangle, as shown for the case of (a) $j_1 = 1/2$ to $j_2 = 1/2$, (b) $j_1 = 1/2$ to $j_2 = 3/2$ and (c) $j_1 = 3/2$ to $j_2 = 3/2$.

surement of a large frequency. Nowadays this can be achieved by laser spectroscopy (Chapter 8) but the near degeneracy of the two $j = 1/2$ levels with $n = 2$ was crucial in Lamb's experiment.[58] Another important feature in that experiment was the metastability of the $2\,\mathrm{S}_{1/2}$ level, whose lifetime was given in Section 2.2.3. That level decays $\sim 10^8$ times more slowly than that of $2\,\mathrm{P}_{1/2}$. In an atomic beam of hydrogen (at room temperature) the atoms have typical velocities of about $3000\,\mathrm{m\,s^{-1}}$ and atoms excited into the 2p configuration travel an average distance of only $5 \times 10^{-6}\,\mathrm{m}$ before decaying with the emission of Lyman-α radiation. In contrast, metastable atoms travel the full length of the apparatus ($\simeq 1\,\mathrm{m}$) and are de-excited when they collide with a detector (or the wall of the vacuum chamber). Hydrogen, and hydrogenic systems, are still used for experimental tests of fundamental theory because their simplicity allows very precise predictions.

[58] Higher shells have smaller shifts between the $j = 1/2$ levels.

2.3.5 Transitions between fine-structure levels

Transitions in hydrogen between the fine-structure levels with principal quantum numbers $n = 2$ and 3 give the components of the Balmer-α line shown in Fig. 2.7; in order of increasing energy, the seven allowed transitions between the levels with different j are as follows:

$$\begin{aligned}
&2\,\mathrm{P}_{3/2} - 3\,\mathrm{S}_{1/2}\,,\\
&2\,\mathrm{P}_{3/2} - 3\,\mathrm{D}_{3/2}\,,\\
&2\,\mathrm{P}_{3/2} - 3\,\mathrm{D}_{5/2}\,,\\
&2\,\mathrm{S}_{1/2} - 3\,\mathrm{P}_{1/2}\,,\\
&2\,\mathrm{P}_{1/2} - 3\,\mathrm{S}_{1/2}\,,\\
&2\,\mathrm{S}_{1/2} - 3\,\mathrm{P}_{3/2}\,,\\
&2\,\mathrm{P}_{1/2} - 3\,\mathrm{D}_{3/2}\,.
\end{aligned}$$

These obey the selection rule $\Delta l = \pm 1$ but an additional rule prevents a transition between $2\,\mathrm{P}_{1/2}$ and $3\,\mathrm{D}_{5/2}$, namely that the change of the total angular momentum quantum number in an electric dipole transition obeys

$$\Delta j = 0, \pm 1. \tag{2.59}$$

This selection rule may be explained by angular momentum conservation (as mentioned in Section 2.2.2). This rule can be expressed in terms of vector addition, as shown in Fig. 2.8; the conservation condition is equivalent to being able to form a triangle from the three vectors representing $\mathbf{j}$ of the initial state, the final state, and a unit vector for the (one unit of) angular momentum carried by the photon. Hence, this selection rule is sometimes referred to as the triangle rule. The projection of $\mathbf{j}$ along the z-axis can change by $\Delta m_j = 0, \pm 1$. (Appendix C gives a summary of all selection rules.)

Further reading

Much of the material covered in this chapter can be found in the introductory quantum mechanics and atomic physics texts listed in the References. For particular topics the following are useful: Segrè (1980) gives an overview of the historical development, and Series (1988) reviews the work on hydrogen, including the important Lamb shift experiment.

Exercises

(2.1) *Angular-momentum eigenfunctions*

(a) Verify that all the eigenfunctions with $l = 1$ are orthogonal to $Y_{0,0}$.

(b) Verify that all the eigenfunctions with $l = 1$ are orthogonal to those with $l = 2$.

(2.2) *Angular-momentum eigenfunctions*

(a) Find the eigenfunction with orbital angular momentum quantum number l and magnetic quantum number $m = l - 1$.

(b) Verify that $Y_{l,l-1}$ is orthogonal to $Y_{l-1,l-1}$.

(2.3) *Radial wavefunctions*
Verify eqn 2.23 for $n = 2$, $l = 1$ by calculating the radial integral (for $Z = 1$).

(2.4) *Hydrogen*
For a hydrogen atom the normalised wavefunction of an electron in the 1s state, assuming a point nucleus, is

$$\psi(r) = \left(\frac{1}{\pi a_0^3}\right)^{1/2} \mathrm{e}^{-r/a_0} ,$$

where a_0 is the Bohr radius. Find an approximate expression for the probability of finding the electron in a small sphere of radius $r_b \ll a_0$ centred on the proton. What is the electronic charge density in this region?

(2.5) *Hydrogen*
The Balmer-α spectral line is observed from a (weak) discharge in a lamp that contains a mixture of hydrogen and deuterium at room temperature. Comment on the feasibility of carrying out an experiment using a Fabry–Perot étalon to resolve (a) the isotope shift, (b) the fine structure and (c) the Lamb shift.

(2.6) *Transitions*
Estimate the lifetime of the excited state in a two-level atom when the transition wavelength is (a) 100 nm and (b) 1000 nm. In what spectral regions do these wavelengths lie?

(2.7) *Selection rules*
By explicit calculation of integrals over θ, for the case of π-polarization only, verify that p to d transitions are allowed, but not s to d.

(2.8) *Spin–orbit interaction*
The spin–orbit interaction splits a single-electron configuration into two levels with total angular momentum quantum numbers $j = l + 1/2$ and $j' = l - 1/2$. Show that this interaction does not shift the mean energy (centre of gravity) of all the states given by $(2j + 1)\, E_j + (2j' + 1)\, E_{j'}$.

(2.9) *Selection rule for the magnetic quantum number*
Show that the angular integrals for σ-transitions contain the factor

$$\int_0^{2\pi} \mathrm{e}^{\mathrm{i}\left(m_{l_1} - m_{l_2} \pm 1\right)\phi}\, \mathrm{d}\phi .$$

Hence derive the selection rule $\Delta m_l = \pm 1$ for this polarization. Similarly, derive the selection rule for the π-transitions.

(2.10) *Transitions*
An atom in a superposition of two states has the wavefunction

$$\Psi\,(t) = A\psi_1\,(\mathbf{r})\, \mathrm{e}^{-\mathrm{i}E_1 t/\hbar} + B\psi_2\,(\mathbf{r})\, \mathrm{e}^{-\mathrm{i}E_2 t/\hbar} .$$

The distribution of electronic charge is given by

$$-e\,|\Psi\,(t)|^2 = -e\big\{\, |A\psi_1|^2 + |B\psi_2|^2 + |2A^* B\psi_1^*\psi_2| \cos\left(\omega_{12} t - \phi\right) \big\}.$$

Part of this oscillates at the (angular) frequency of the transition $\omega_{12} = \omega_2 - \omega_1 = (E_2 - E_1)\,/\hbar$.

(a) A hydrogen atom is in a superposition of the 1s ground state, $\psi_1 = R_{1,0}\,(r)\, Y_{0,0}\,(\theta, \phi)$, and the $m_l = 0$ state of the 2p configuration, $\psi_2 = R_{2,1}\,(r)\, Y_{1,0}\,(\theta, \phi)$; $A \simeq 0.995$ and $B = 0.1$ (so the term containing B^2 can be ignored). Sketch the form of the charge distribution for one cycle of oscillation.

(b) The atom in a superposition state may have an oscillating electric dipole moment

$$-e\mathbf{D}\,(t) = -e\,\langle \Psi^*\,(t)\, \mathbf{r}\Psi\,(t)\rangle .$$

What are the conditions on ψ_1 and ψ_2 for which $\mathbf{D}\,(t) \neq 0$.

(c) Show that an atom in a superposition of the same states as in part (a) has a dipole moment of

$$-e\mathbf{D}\,(t) = -e\,|2A^* B|\, \mathcal{I}_{\mathrm{ang}} \times \left\{ \int r R_{2,1}(r) R_{1,0}(r) r^2\, \mathrm{d}r \right\} \cos(\omega_{12} t)\, \widehat{\mathbf{e}}_z ,$$

where $\mathcal{I}_{\mathrm{ang}}$ is an integral with respect to θ and ϕ. Calculate the amplitude of this dipole, in units of ea_0, for $A = B = 1/\sqrt{2}$.

(d) A hydrogen atom is in a superposition of the 1s ground state and the $m_l = 1$ state of the 2p configuration, $\psi_2 = R_{2,1}\,(r)\, Y_{1,1}\,(\theta, \phi)$. Sketch the form of the charge distribution at various points in its cycle of oscillation.

(e) Comment on the relationship between the time dependence of the charge distributions sketched in this exercise and the motion of the electron in the classical model of the Zeeman effect (Section 1.8).

(2.11) *Angular eigenfunctions*
We shall find the angular momentum eigenfunctions using ladder operators, by assuming that for some value of l there is a maximum value of the magnetic quantum number $m_{\max}$. For this case $Y_{l,m_{\max}} \propto \Theta(\theta)\mathrm{e}^{\mathrm{i}m_{\max}\phi}$ and the function $\Theta(\theta)$ can be found from

$$l_+ \Theta(\theta) \exp\left(\mathrm{i}m_{\max}\phi\right) = 0 .$$

(a) Show that $\Theta(\theta)$ satisfies the equation

$$\frac{1}{\Theta(\theta)}\frac{\partial\Theta(\theta)}{\partial\theta} = m_{\max}\frac{\cos\theta}{\sin\theta}\,.$$

(b) Find the solution of the equation for $\Theta(\theta)$. (Both sides have the form $f'(\theta)/f(\theta)$ whose integral is $\ln\{f(\theta)\}$.) By substituting this solution into eqn 2.5 to show that $b = m_{\max}(m_{\max}+1)$, or otherwise, obtain eqn 2.10.

(2.12) *Parity and selection rules*
Show that eqn 2.42 implies that $l_2 - l_1$ is odd. Hence, or otherwise, prove that $\mathcal{I}_{\mathrm{ang}}$ is zero unless the initial and final states have opposite parity.

(2.13) *Selection rules in hydrogen*
Hydrogen atoms are excited (by a pulse of laser light that drives a multi-photon process) to a specific configuration and the subsequent spontaneous emission is resolved using a spectrograph. Infrared and visible spectral lines are detected *only* at the wavelengths $4.05\,\mu$m, $1.87\,\mu$m and $0.656\,\mu$m. Explain these observations and give the values of n and l for the configurations involved in these transitions.

Web site:

`http://www.physics.ox.ac.uk/users/foot`

This site has answers to some of the exercises, corrections and other supplementary information.